

420-301-VA: Web Services

Lab 3: HTTP Fundamentals and Web Service Clients

Part 1

```
1  import java.io.*;
2  import java.net.*;
3
4  public class RawHttpClient {
5      Run | Debug | Run main | Debug main
6      public static void main(String[] args) {
7          String host = "example.com";
8          int port = 80;
9
10         try(Socket socket= new Socket(host, port)){
11             BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
12             PrintWriter out = new PrintWriter(socket.getOutputStream(), autoFlush: true);
13
14             out.print(s: "GET / HTTP/1.1\r\n");
15             out.print("Host: " + host + "\r\n");
16             out.print(s: "Connection: close\r\n");
17             out.print(s: "\r\n");
18             out.flush();
19
20             String line;
21             while((line = in.readLine()) != null){
22                 System.out.println(line);
23             }
24         } catch (IOException e){
25             System.out.println("Error: " + e.getMessage());
26         }
27     }
28 }
```

```
[Running] cd "c:\Users\efren\Documents\Vanier\Semester 7\Web Services\Lab 3\Exercise 1\" && javac RawHttpClient.java && java RawHttpClient
HTTP/1.1 200 OK
Date: Fri, 11 Sep 2026 01:31:22 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Last-Modified: Thu, 10 Sep 2026 20:59:30 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 6749
cf-cache-status: HIT
CF-RAY: a392e9988da4a29a-YUL

22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width,
initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</
style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a
href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

0

[Done] exited with code=0 in 0.563 seconds
```

Part 2

```
Exercise 1 > HttpInspector.java > Language Support for Java(TM) by Red Hat > HttpInspector > main(String[])
1 import java.io.*;
2 import java.net.*;
3
4 public class HttpInspector {
5     public static void main(String[] args) {
6         String host = "example.com";
7         int port = 80;
8
9         try(Socket socket= new Socket(host, port)){
10             BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
11             PrintWriter out = new PrintWriter(socket.getOutputStream(), autoFlush: true);
12
13             out.print(s: "GET /this-page-does-not-exist.html HTTP/1.1\r\n"); // Single backslash!
14             out.print(s: "Host: " + host + "\r\n");
15             out.print(s: "Connection: close\r\n");
16             out.print(s: "\r\n");
17             out.Flush();
18
19             String statusLine = in.readLine();
20             System.out.println("STATUS: " + statusLine);
21
22             System.out.println(s: "--- HEADERS ---");
23             String headerLine;
24             while ((headerLine = in.readLine()) != null && !headerLine.isEmpty()) {
25                 System.out.println(headerLine);
26             }
27
28             System.out.println(s: "--- BODY ---");
29             String bodyLine;
30             while((bodyLine = in.readLine()) != null) {
31                 System.out.println(bodyLine);
32             }
33         } catch(IOException e){
34             System.out.println("Error: " + e.getMessage());
35         }
36     }
37 }
38 }
```

```
[Running] cd "c:\Users\efren\Documents\Vanier\Semester 7\Web Services\Lab 3\Exercise 1\" && javac HttpInspector.java && java HttpInspector
STATUS: HTTP/1.1 404 Not Found
--- HEADERS ---
Date: Fri, 11 Sep 2026 01:47:23 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Age: 7553
cf-cache-status: HIT
CF-RAY: a393010dfdc2be-YUL
--- BODY ---
22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width,
initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</
style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a
href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

0

[Done] exited with code=0 in 0.575 seconds
```

```
[Running] cd "c:\Users\efren\Documents\Vanier\Semester 7\Web Services\Lab 3\Exercise 1\" && javac HttpInspector.java && java HttpInspector
STATUS: HTTP/1.1 301 Moved Permanently
--- HEADERS ---
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-J4Hd1LWL70eMoTBnhXrSyA' 'strict-dynamic' 'report-sample' 'unsafe-eval'
'unsafe-inline' https: http:report-uri https://csp.withgoogle.com/csp/gws/other-hp
Date: Fri, 11 Sep 2026 01:51:31 GMT
Expires: Sun, 11 Oct 2026 01:51:31 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN
Connection: close
--- BODY ---
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>

[Done] exited with code=0 in 0.675 seconds
```

Thinking Questions

- Statelessness:

By caching your data. For example, when using google.com as a host, we can see that the caching data has an expiration date of 30 days.

- HTTPS (Security):

Because the connection immediately closes once it is made. This is due to the threeway handshake (SYN SYNACK ACK). In addition, a standard socket is unencrypted, so it is considered invalid

Part 3

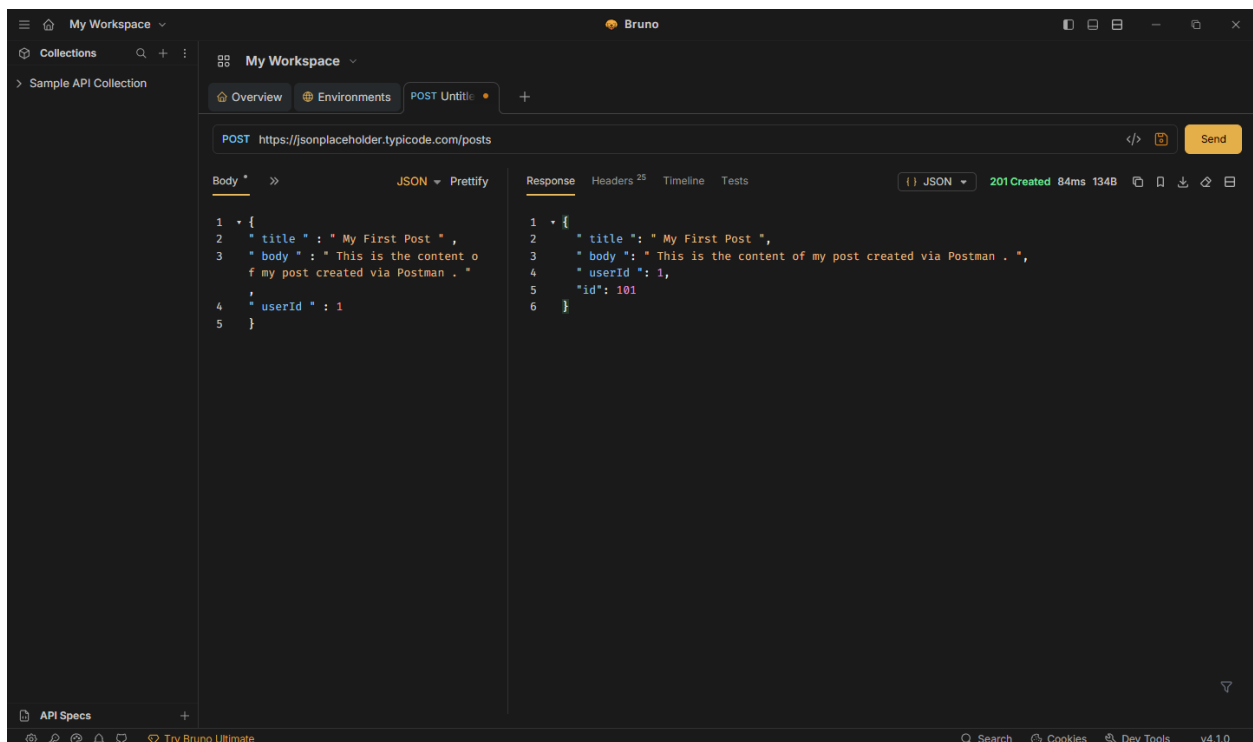
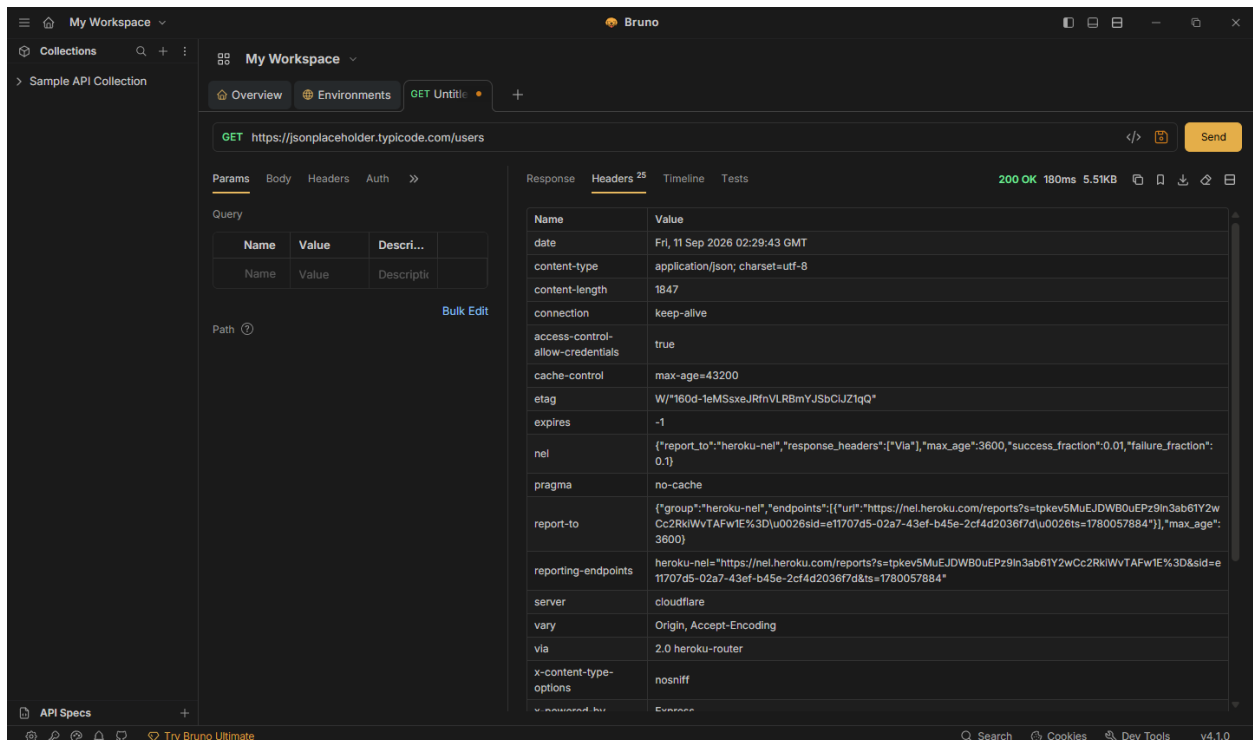
```
C:\Users\efren>curl http://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operation s.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

C:\Users\efren>curl -I http://example.com
HTTP/1.1 200 OK
Date: Fri, 11 Sep 2026 02:05:54 GMT
Content-Type: text/html
Connection: keep-alive
Server: cloudflare
Last-Modified: Thu, 10 Sep 2026 20:59:30 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 8821
cf-cache-status: HIT
CF-RAY: a3931c2fbc71a286-YUL
```

```
C:\Users\efren>curl -v http://example.com
* Host example.com:80 was resolved.
* IPv6: 2606:4700:10::ac42:93f3, 2606:4700:10::6814:179a
* IPv4: 172.66.147.243, 104.20.23.154
* Trying [2606:4700:10::ac42:93f3]:80...
* Established connection to example.com (2606:4700:10::ac42:93f3 port 80) from 2001:4958:3ff8:7e01:f50e:ff58:3b29:ff42 port 51428
* using HTTP/1.x
> GET / HTTP/1.1
> Host: example.com
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 11 Sep 2026 02:06:33 GMT
< Content-Type: text/html
< Transfer-Encoding: chunked
< Connection: keep-alive
< Server: cloudflare
< Last-Modified: Thu, 10 Sep 2026 20:59:30 GMT
< Allow: GET, HEAD
< Accept-Ranges: bytes
< Age: 8860
< cf-cache-status: HIT
< CF-RAY: a3931d22bdadfaf-YUL
<
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operation s.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
* Connection #0 to host example.com:80 left intact
```

Part 4

For this part im using Bruno rather than Postman and Insomnia



Thinking Question

Based on the raw socket work you did in Part 1, what exactly did Postman send behind the scenes when you clicked “Send” for the POST request? What Headers must it have

added automatically so the server knew it was receiving JSON data?

Postman (or Bruno in my case) sends an HTTP request. It added:

- Content-Type: Format of the data (Tells the server the data is in json)
- Authorization: Credential to let server know who you are
- User_agent: the client who is making the request